Epic Games Store Giveaways: Causal Effect on Steam Player Engagement

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Motivation & Research Question

Motivating example

- Epic Games Store (EGS) has given away over 400 games since 2018 — its primary differentiation lever against Steam
- Standard theory predicts a zero-price substitute on a rival platform diverts players *away* from Steam
- But in January 2026, New Blood CEO Dave Oshry reported that giving away *Blood West* on EGS caused Steam sales to surge $\sim$200% the same day

Research question

Does an EGS free giveaway *increase* or *decrease* Steam player engagement for the same game — and how does this evolve over time?

Two competing mechanisms

Advertising: Giveaway generates media buzz $\rightarrow$ players flock to Steam (primary discovery/community layer) $\rightarrow$ positive short-run spillover

Substitution: Free copy on Epic gradually diverts playtime away from Steam $\rightarrow$ negative medium-run effect

Both may operate, but at different horizons.

Data

EGS Giveaway History

- Kaggle dataset (Dongre, 2025): every free game on EGS Dec 2018–2025
- 541 events → 471 unique titles; 70 repeat appearances
- 63 titles given away more than once

Steam Player Data

- Monthly average concurrent players scraped from SteamCharts for each matched game
- Panel: July 2012 – February 2026
- Outcome: $\ln(\texttt{avg_players} + 1)$

Final Panel

Games in final panel	446
Game-month observations	38,642
Median game months of data	88
Mean avg. players/month	1,584
(median)	65
Mean $\ln(\texttt{avg_players} + 1)$	4.43

Player counts are heavily right-skewed — 101 games have pre-treatment median < 10 players. Log transformation compresses the tail. **Matching:** Fuzzy string matching + manual review; 8 titles confirmed absent from Steam

Identification Strategy

Design: Staggered DiD Event Study

- Treatment = game i's first EGS giveaway, staggered across 80 calendar months (Dec 2018 – late 2025)
- **Control group:** not-yet-treated games — games awaiting future giveaways serve as controls for currently treated games
- No external never-treated group needed

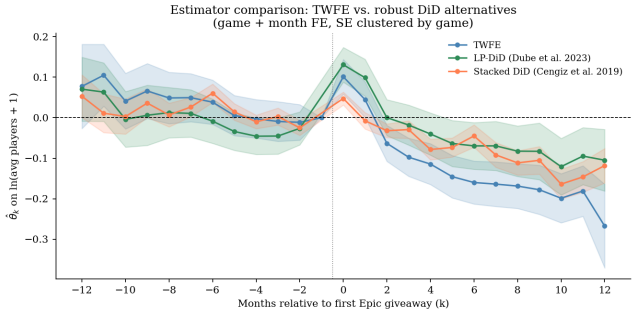
Key assumption (parallel trends)
In absence of treatment, log player trajectories would trend identically after conditioning on game & calendar-month FE. Pre-period coefficients ($k \in [-12, -2]$) trace a downward slope ($\chi^2 = 21.77$, $p = 0.026$) — treated games were *already declining* before giveaway (lifecycle selection). High-traffic subsample passes the test ($p = 0.589$).

Four Estimators (for robustness)

1. **TWFE** — primary specification; game + calendar-month FE; event window $k \in [-12, 12]$
2. **LP-DiD** (Dubé et al., 2023) — local projections using only not-yet-treated controls at each k; robust to heterogeneous effects
3. **Stacked DiD** (Cengiz et al., 2019) — cohort-specific balanced panels; avoids negative-weight bias
4. **CS-ATT** (Callaway & Sant'Anna, 2021) — cohort-specific ATTs; most conservative

Randomization placebo: 200 permutations of treatment dates to validate the $k = 0$ spike

Results: Two-Phase Dynamic



TWFE, LP-DiD, and Stacked DiD. Game + calendar-month FE. $N = 38{,}641$; 446 games. Reference: $k = -1$.

Phase 1 — Advertising spike

- $+5$ to $+14\%$ more Steam players at $k = 0$ across all estimators
- Stacked DiD ($+4.7\%$) is lower bound; LP-DiD ($+13.1\%$) upper bound
- Placebo $p < 0.001$; actual estimate is $2.7\times$ the 95th percentile of null

Phase 2 — Medium-run decline

- All estimators turn negative by $k = 2$; significant from $k = 5$ onward
- -10 to -23% below baseline by $k = 12$ across estimators
- Primarily pre-existing lifecycle decline

Robustness: Why Do Estimators Differ?

Control-group composition drives the ordering

TWFE mixes in already-treated controls that have passed the spike and partially stabilized — less actively declining, so less upward pull at $k = 0$ and more negative at large k.

LP-DiD uses *not-yet-treated* controls — games still approaching their own giveaway, in the steepest phase of pre-treatment lifecycle decline. Subtracting actively declining controls inflates LP-DiD upward at $k = 0$, and attenuates it toward zero at large k.

Stacked DiD balances cohort-specific panels, neutralizing the composition effect $\Rightarrow$ most conservative at $k = 0$; reversal appears by $k = 1$ rather than $k = 2$.

k	TWFE	LP-DiD	Stacked
0	$+0.101^{***}$	$+0.131^{***}$	$+0.047^{***}$
1	$+0.045^{*}$	$+0.099^{***}$	-0.008
2	-0.063^{**}	0.000	-0.032^{***}
5	-0.145^{***}	-0.064^{*}	-0.073^{***}
10	-0.199^{***}	-0.121^{***}	-0.164^{***}
12	-0.267^{***}	-0.105^{**}	-0.118^{***}

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *SE clustered by game.*

Same qualitative two-phase pattern across all three; quantitative spread reflects control-group composition, not a contradiction.

Takeaways, Caveats & Future Work

Main takeaway: Both mechanisms operate at different horizons — advertising spike (+5–14%) then lifecycle decline (−10–23%); LP-DiD −10–12% is an upper bound on true causal substitution.

Developers: Fee + Steam spike are both positive; player migration to Epic is not a developer cost (12% vs. 30% cut).

Steam: Net positive — spike accrues to Steam; decline is mostly pre-existing lifecycle, not loss of engaged users.

Epic: Open question — measurable effects accrue to Steam; whether EGS user acquisition offsets the licensing cost is unknown.

Caveats

- **Lifecycle selection** — treated games already declining pre-giveaway
- **Players $\neq$ purchases** — concurrent counts, not sales or revenue
- **Genre heterogeneity** — pooled estimate mixes single-player and multiplayer

Future work

- Single-player vs. multiplayer heterogeneity
- Steam sales volume effects
- EGS-side engagement/sales data